

Task/Case Study 5

Business Recommendations

For NovaByte Solutions Leadership

Prepared by Satyabrata Sahoo

Career 247 — Business Analytics with GenAI

Context

NovaByte's leadership asked for a root-cause diagnosis of a revenue slowdown that began in Q3-2024, despite a healthy pipeline, rising marketing spend, and higher approved budgets. Four case studies' worth of cleaning, SQL investigation, and statistical testing point to a clear conclusion: the slowdown is not a single, simple problem. It's several smaller, specific issues — in pipeline volume, marketing spend allocation, support responsiveness, and financial process — that individually look survivable but compound into the revenue picture leadership is seeing. The seven recommendations below are ranked by evidence strength and business impact, each tied to a specific, traceable data point from the analysis rather than a general impression.

Recommendation 1: Investigate Pipeline Volume, Not Win Rate or Deal Speed

OWNER: Sales

Finding: Closed Won revenue fell 33.5% from Q1-2023 (₹2.33 crore) to Q1-2025 (₹1.55 crore). Formal hypothesis testing ruled out two of the most obvious explanations: win rate does not differ significantly between digital and offline lead sources ($p=0.848$), and deal closure time has not significantly changed over the last year ($p=0.501$).

Why It Matters: If deals aren't converting worse and aren't taking longer, the revenue decline has to be coming from somewhere upstream — fewer deals entering the pipeline, or smaller average deal sizes, neither of which this analysis was scoped to test directly. Continuing to focus on conversion-rate fixes (sales training, process speed) would very likely fail to move revenue, because conversion isn't where the problem lives.

Recommended Action: Sales operations should build a quarterly “new pipeline created” trend (deal count and total value, by Created_Date) alongside the existing Closed Won trend, and compare the two. If new pipeline volume is also declining, the root cause is a prospecting or lead-generation problem, not a sales execution one — which changes who should own the fix.

Recommendation 2: Fix SLA Compliance for Critical-Priority Tickets First

OWNER: Support

Finding: Overall SLA compliance is weak across the board (35–41%), but it is worst for Critical-priority tickets specifically (34.6%) — the exact opposite of where compliance should be strongest, since Critical tickets carry the tightest SLA targets and the highest stakes.

Why It Matters: A blanket “improve SLA compliance” initiative would spread effort evenly across all priority levels and likely leave the highest-risk tickets exactly where they are today. The client relationships most exposed to a bad support experience are being served worst, not best.

Recommended Action: Support leadership should audit staffing and escalation routing specifically for Critical-priority tickets — confirm they're reaching senior agents immediately rather than sitting in a general queue — before investing in any broader SLA improvement program.

Recommendation 3: Build a Proactive Churn Early-Warning Trigger from Ticket Volume

OWNER: Customer Success / Account Management

Finding: Of the top 4 clients by support ticket volume, 3 have already moved to Churned status — including one with a ₹5,00,000 contract, among the largest in the dataset. This pattern held regardless of contract size.

Why It Matters: Right now, high ticket volume is only visible in Support's own systems — it isn't connected to account health monitoring or renewal risk flagging elsewhere in the business. By the time churn shows up in revenue or contract-renewal reporting, it's too late to intervene.

Recommended Action: Customer Success (or Account Management, if Support doesn't own renewals) should add a simple threshold alert — e.g., any client crossing 15+ tickets in a rolling 90-day window triggers a manual account-health review — rather than waiting for churn to surface through other channels.

Recommendation 4: Shift Marketing Budget Away From Email, Toward Google Ads and Trade Show

OWNER: Marketing

Finding: Email has the smallest total spend of any of the 8 tracked channels (₹1.05 crore) yet by far the worst customer acquisition cost (₹95,675 — over 6x Google Ads' ₹14,957). Meanwhile Trade Show tops closed-deal value (₹2.87 crore) despite not being a leader in either spend or measured ROI, and Google Ads combines the largest spend with the lowest CAC of any channel.

Why It Matters: Marketing dollars are currently being spent least efficiently on the channel that needs the most spend to produce a customer, while a channel already converting well (Trade Show) isn't being scaled, and the clear efficiency leader (Google Ads) has room to absorb more budget without a proportional cost increase.

Recommended Action: Reallocate a defined share of Email's budget (a pilot of 20–30% is a reasonable starting point) into Google Ads and Trade Show-style channels next quarter, and re-measure CAC and closed-deal value attribution after one full quarter to confirm the shift is working before going further.

Recommendation 5: Give Cold Outreach a Formal Channel Entry in Marketing's Funnel Tracking

OWNER: Sales / Marketing (joint)

Finding: Cold Outreach — a lead source with zero tracked marketing spend — generated ₹2.07 crore in closed-deal value, outperforming both LinkedIn and Referral in raw closed value. It currently has no corresponding entry in Marketing_Campaigns, meaning this revenue is invisible to Marketing's own funnel and ROI reporting.

Why It Matters: A meaningful share of what's actually working is currently unmeasured and unmanaged. Sales reps generating this revenue aren't being recognized in the metrics leadership uses to evaluate channel performance, and there's no way to know if this effort could scale with any investment behind it at all.

Recommended Action: Sales and Marketing should jointly define what “counts” as Cold Outreach effort (rep hours, tooling cost, or a nominal budget line) and add it as a tracked entry in the campaign data, even if the true cost is currently near zero — so its contribution is visible on the same terms as every other channel.

Recommendation 6: Clear the ₹8.99 Crore Expense Approval Backlog

OWNER: Finance

Finding: 181 expense records above ₹2,00,000 — totaling roughly ₹8.99 crore — are currently sitting in Pending or Rejected approval status (120 Pending, 61 Rejected), with no visible resolution timeline in the data.

Why It Matters: Money in unresolved approval limbo is money leadership can't confidently plan around, and a backlog this size at this value threshold suggests either an under-resourced approval process or a policy gap in how large expenses get escalated and decided.

Recommended Action: Finance should set a defined approval SLA for any expense above ₹2,00,000 (10 business days is a reasonable starting benchmark) and report backlog age and total value monthly until the current 181-item backlog is cleared.

Recommendation 7: Review Why the East Region Converts Marketing Spend So Poorly

OWNER: Marketing / Regional
Leadership

Finding: East carries the second-highest marketing spend of any region (₹2.19 crore) but the lowest revenue per client (₹1,53,419) of all four regions — the weakest return on marketing investment anywhere in the business. South, by contrast, achieves strong revenue per client with the leanest marketing spend and the best SLA compliance of any region.

Why It Matters: Without correcting for this, East will keep absorbing a disproportionate share of next quarter's marketing budget for a below-average return, while a genuinely more efficient approach (South's) goes unexamined and unreplicated elsewhere.

Recommended Action: Marketing and regional sales leadership should run a side-by-side review of East's channel mix and lead quality against South's before committing next quarter's regional budget allocation — the goal is to understand what South is doing differently, not simply to cut East's spend.

Summary Table

#	Recommendation	Owner
1	Investigate pipeline volume, not win rate	Sales
2	Fix SLA compliance for Critical tickets first	Support
3	Build a churn early-warning trigger	Customer Success
4	Shift budget from Email to Google Ads/Trade Show	Marketing
5	Formalize Cold Outreach tracking	Sales / Marketing
6	Clear the ₹8.99cr expense approval backlog	Finance
7	Review East region's marketing efficiency	Marketing / Regional